

AI SMART HOME AUTOMATION SYSTEM USING ESP32

Abstract

This project presents an AI-integrated Smart Home Automation System using ESP32, Blynk IoT, and Sinric Pro. The system enables users to control electrical appliances remotely using a mobile application and Google Assistant voice commands. The project demonstrates practical IoT implementation, cloud communication, and wireless automation for smart home applications.

Objectives

- To automate household appliances using ESP32.
- To integrate Blynk IoT for mobile control.
- To implement Google Assistant voice control using Sinric Pro.
- To provide wireless smart home automation using WiFi.

Components Used

- ESP32 Development Board
- 3-Channel Relay Module
- Bulbs
- Fan
- Jumper Wires
- Breadboard
- WiFi Network

Software Used

- Arduino IDE
- Blynk IoT
- Sinric Pro
- Google Home

Working Principle

The ESP32 connects to the WiFi network and communicates with Blynk IoT and Sinric Pro cloud services. Users can control connected appliances using the Blynk mobile application or Google Assistant voice commands. The ESP32 receives commands from the cloud platform and switches the relay module accordingly.

Features

- Mobile App Control
- Voice Assistant Integration

- Real-Time Relay Switching
- Wireless Automation
- IoT Cloud Integration
- Smart Appliance Control

Applications

- Smart Homes
- Voice-Controlled Automation
- Energy Saving Systems
- Remote Appliance Monitoring
- IoT Educational Projects

Advantages

- Easy to use
- Remote access capability
- Low power consumption
- Cost-effective implementation
- Real-time operation

Future Enhancements

- Motion Sensor Integration
- Energy Monitoring
- ESP32-CAM Security System
- Temperature-Based Automation
- AI-Based Smart Scheduling

Conclusion

The AI Smart Home Automation System successfully demonstrates the integration of IoT, cloud platforms, and embedded systems for modern smart home applications. The project provides a reliable, scalable, and user-friendly solution for wireless appliance control.